1 + 8 = 9: small enough to check by hand

a = 1	b = 8	c = 9
no prime support	2^3	3^2

$$R = 2 \cdot 3 = 6 \qquad c/R = 3/2 \qquad Q = 12$$

$$q = \log 9 / \log 6 \approx 1.226294$$

Primitive means gcd(1,8) = 1. The prime sectors are disjoint.